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AI-Based Traffic Violation Detection System

ABSTRACT

Traffic violations are a significant contributor to road accidents, congestion, and law enforcement inefficiencies.

This project presents an AI-Based Traffic Violation Detection System that automates the detection and reporting of traffic violations using **Computer Vision (CV)**, **Deep Learning (DL)**, and **IoT devices**. The system captures live video feeds from traffic cameras, detects violations such as red light jumping, overspeeding, and illegal parking, and automatically logs violator information including vehicle number plates. Evaluation on simulated traffic scenarios demonstrates high detection accuracy and a significant reduction in manual monitoring effort.

The report details requirements, design, implementation, evaluation, risks, and future work.

Assignment 1: Preliminary Investigation and its Activities

1.1 Problem Identification and Definition

Traffic authorities face difficulties in monitoring and enforcing traffic laws manually due to the high volume of vehicles and limited personnel.

Traditional systems rely on human observation, which is time-consuming, error-prone, and inefficient for continuous monitoring. With the growing number of vehicles on roads, manual traffic violation detection leads to missed violations and delayed enforcement.

Additionally, it increases the chances of corruption and inconsistent penalty enforcement.AI-based systems offer automated, accurate, and real-time traffic violation detection. By integrating computer vision and machine learning algorithms with real-time camera feeds, the system can analyze vehicle behavior, detect violations, and issue tickets automatically.

1.2 Problem Description

High vehicle density: Thousands of vehicles pass through intersections, making manual observation impractical.

Human error: Manual monitoring leads to missed violations or false reporting due to oversight or fatigue.

Lack of real-time enforcement: Delayed reporting hampers timely penalty enforcement.

Insufficient data collection: Manual systems lack structured data for trend analysis and urban planning.

1.3 Fact-Finding Techniques

Interviews – Discussions with traffic police officers to understand current limitations and workflow.

Questionnaires/Surveys – Gathering feedback from commuters and traffic officers regarding common violation issues.

Observation – Studying traffic flow and violation patterns at intersections.- Document Analysis – Reviewing accident reports and violation records.

Prototyping – Building a small-scale demo system to test violation detection accuracy.

1.4 Drawbacks of Existing System

- Reliance on manual observation leads to missed violations.

- High operational cost due to personnel requirements.

- Inconsistent enforcement and chances of human bias or error.

- Limited ability to store and analyze historical violation data.

1.5 Scope of the Proposed System

- **For Traffic Authorities**:

  - Automated detection of red light jumping, speeding, illegal parking.

  - Real-time capture of violation images/videos and vehicle number plate recognition.

  - Generation of violation reports and automated fine issuance.

- **For Commuters**:

  - Automated notification of violations and penalty status.

- **For Government**:

  - Data analytics on traffic violation trends to inform policy. The system targets integration with city traffic management infrastructure, smart traffic lights, and government portals for fine collection and data storage.

1.6 Feasibility Study

- **Technical Feasibility**: Possible using computer vision models, YOLO for object detection, OCR for number plate reading, and IoT infrastructure.

- **Economic Feasibility**: Reduces operational cost compared to manual enforcement, justified by lower long-term expenses.

- **Operational Feasibility**: User-friendly interfaces for traffic authorities to monitor and manage violations.

- **Schedule Feasibility**: Project development achievable in 6–8 months with a small team of developers and traffic domain experts.

Assignment 2: Specifications and Requirements

2.1 Data Requirements of the System

- **Input Data**:

  - Live video feed from traffic cameras.

  - Configurable violation rules (e.g., speed limits, traffic light status).

- **Stored Data**:

  - Violation records (timestamp, vehicle type, number plate).

  - Camera and sensor configurations.

- **Output Data**:

  - Violation reports with evidence.

  - Alerts to responsible authorities.

2.2 Identify End Users of the System

- **Traffic Police** – Monitor and manage violations automatically.

- **City Administration** – Analyze data for urban planning and policy decisions.

- **Commuters** – Receive automated violation notices.

2.3 Input Data to the System

-Real-time video streams from traffic cameras.

-Traffic signal status data.

-Configuration parameters for speed limits, monitored zones, etc.

2.4 Output Information from the System

- Violation reports with images, time, and vehicle number.

- Real-time alerts.

- Historical trend reports.

- Fine issuance records.

2.5 Functional / Non-Functional / Processing Requirements

A. Functional Requirements

- **Video Stream Input** – Real-time processing of video from cameras.

- **Violation Detection** – Detect specific traffic rule violations using trained models.

- **Number Plate Recognition** – Use OCR to extract vehicle number plates.

- **Report Generation** – Auto-generate detailed violation reports.

- **Notification System** – Send violation notices via SMS or email.

- **Data Storage** – Store violation data securely.

B. Non-Functional Requirements

- **Performance** – Process video frames at ≥ 15 fps for real-time monitoring.

- **Scalability** – Support multiple camera feeds simultaneously.

- **Security & Privacy** – Encrypt personal data and violation reports.

- **Reliability** – High accuracy in violation detection (> 95%).

- **Usability** – Easy-to-use dashboard for traffic officers.

- **Portability** – Accessible on desktops and tablets.

- **Maintainability** – Simple process to update detection models and rules.

C. Processing Requirements

-**Capture Input** – Receive live video feed from cameras and traffic signal status.

-**Frame Extraction** – Break video into frames for analysis.

-**Violation Detection** – Apply AI/Computer Vision models (e.g., YOLO) to detect rule violations.

-**Number Plate Recognition** – Use OCR (Tesseract) to read license plates.

-**Database Update** – Store violation record with timestamp, vehicle number, and evidence image.

-**Report Generation** – Create structured violation reports.

-**Notification** – Automatically send alerts to violators and traffic authorities.

-**Analytics** – Generate trend reports for city administration and policymakers.

Assignment 3: Database Design

3.1 Identify Entities and Attributes

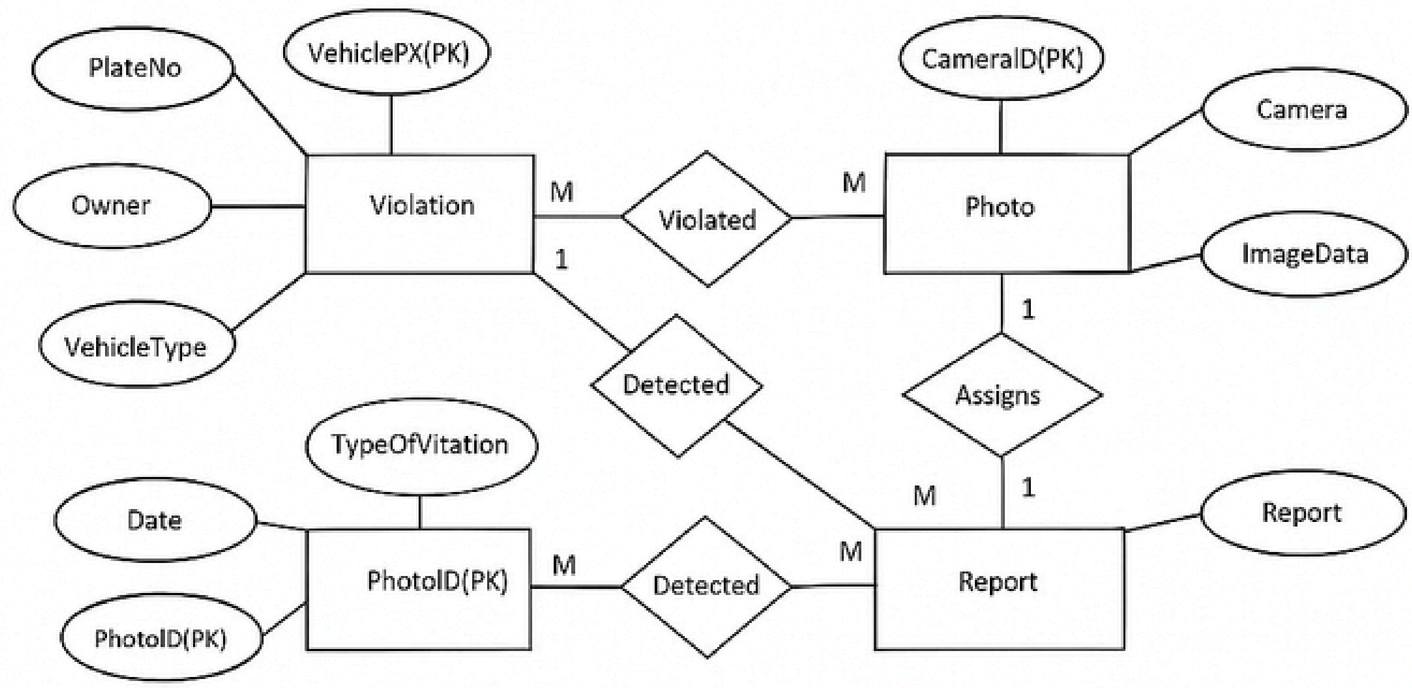
1. Violation : (ViolationID, Timestamp, VehicleNumber, ViolationType, ImagePath).

2. Vehicle : (VehicleID, NumberPlate, OwnerName, VehicleType).

3. Camera : (CameraID, Location, IPAddress, Status).

4. Configuration : (ConfigID, SpeedLimit, MonitoredZones, SignalRules).

3.2 E-R Diagram



3.3 Tables, Fields, Relationships

The database design for the system includes the following main tables:

1. Violation Table

Field Name	Data type	Description
Violation_ID (PK)	INT	Unique ID
Timestamp	DATETIME	Time of Violation
Vehicle_Number	VARCHAR	Vehicle Plate
Violation_Type	VARCHAR	Red Light, Over Speed, etc.
Image_Path	VARCHAR	Path to Evidence Image.

Relationship: Many **violations** (different records) are associated with one specific point in time each.

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2. Vehicle Table

Field Name	Data Type	Description
Vehicle_ID (PK)	INT	Unique ID
Number_Plate	VARCHAR	Vehicle Plate
Owner_Name	VARCHAR	Owner's Name
Vehicle_Type	VARCHAR	Car, Truck, etc.

Relationship: Each **Violation** happened at exactly **one timestamp** (Many-to-One concept, stored as an attribute).

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3. Camera Table

Field Name	Data Type	Description
Camera_ID (PK)	INT	Unique ID
Location	VARCHAR	Installation Location
IP_Address	VARCHAR	Network Address
Status	VARCHAR	Active/Inactive

Relationship: A single camera records multiple violations over time, but each violation is captured by exactly one camera.

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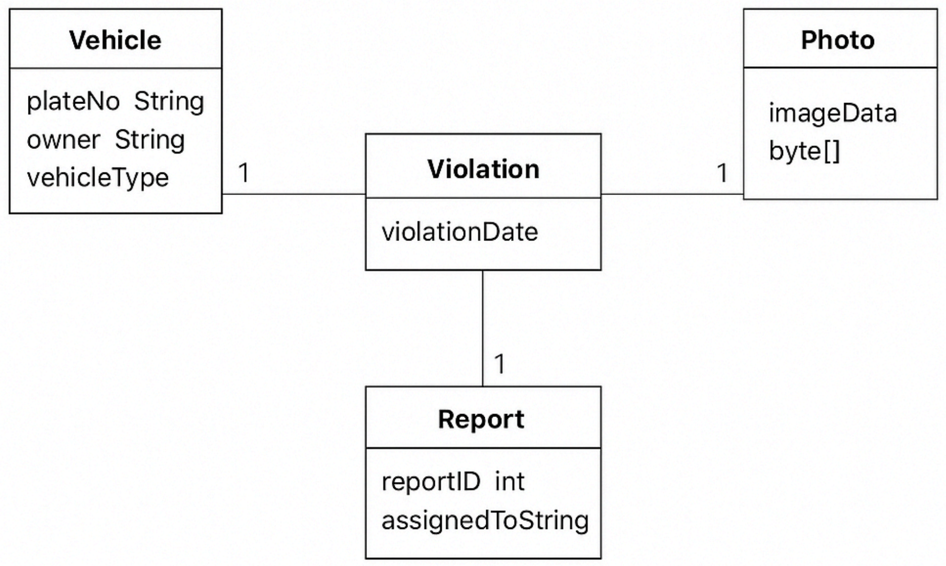
4. Configuration Table

Field Name	Data Type	Description
Config_ID (PK)	INT	Unique ID
Speed_Limit	DECIMAL	Speed light value.
Monitored_Zones	TEXT	Coordinates of zones.
Signal_Rules	TEXT	Rules for red light enforcement

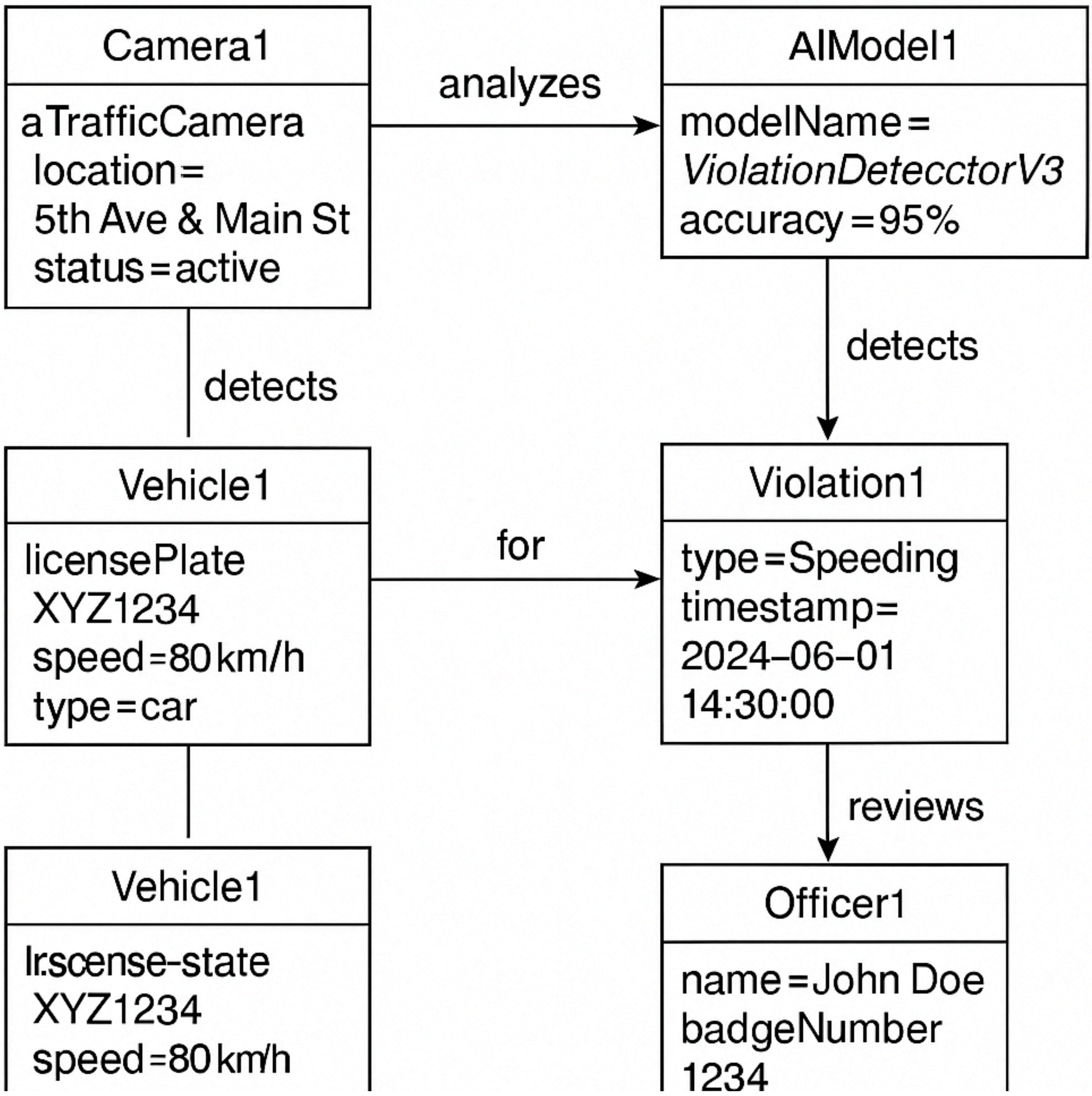
Relationship: If multiple cameras share the same configuration → **Many Cameras → One Configuration (Many-to-One)**

Assignment 4: System Design

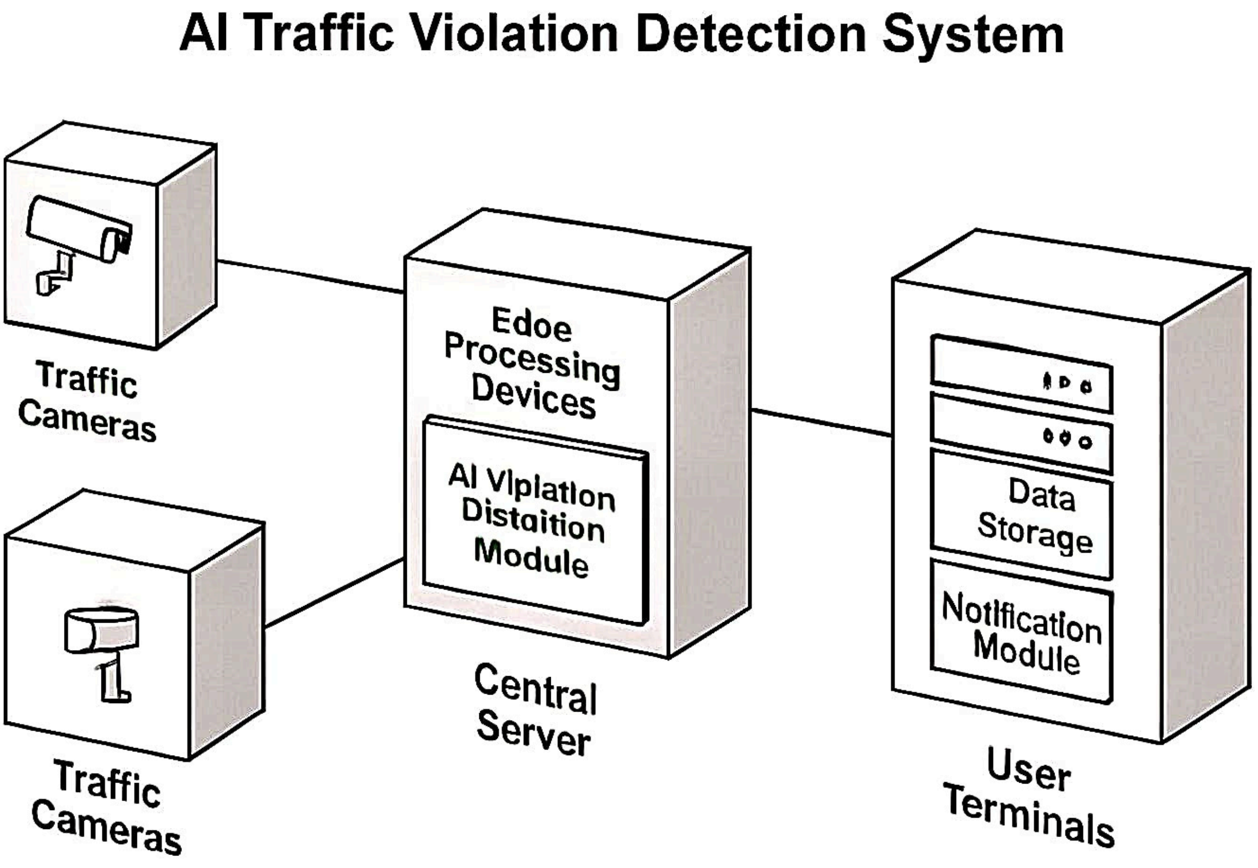
4.1 Class Diagram



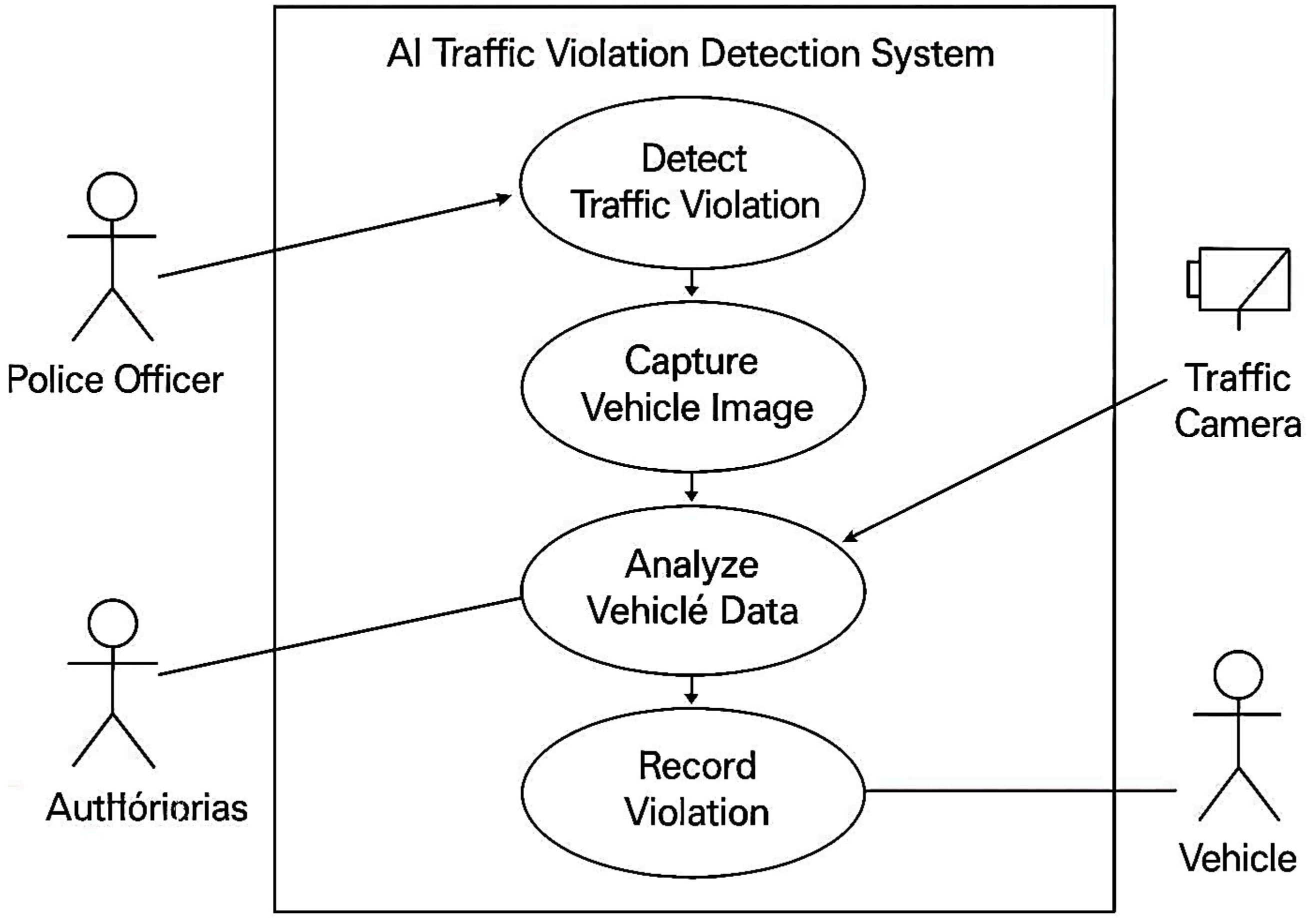
4.2 Object Diagram



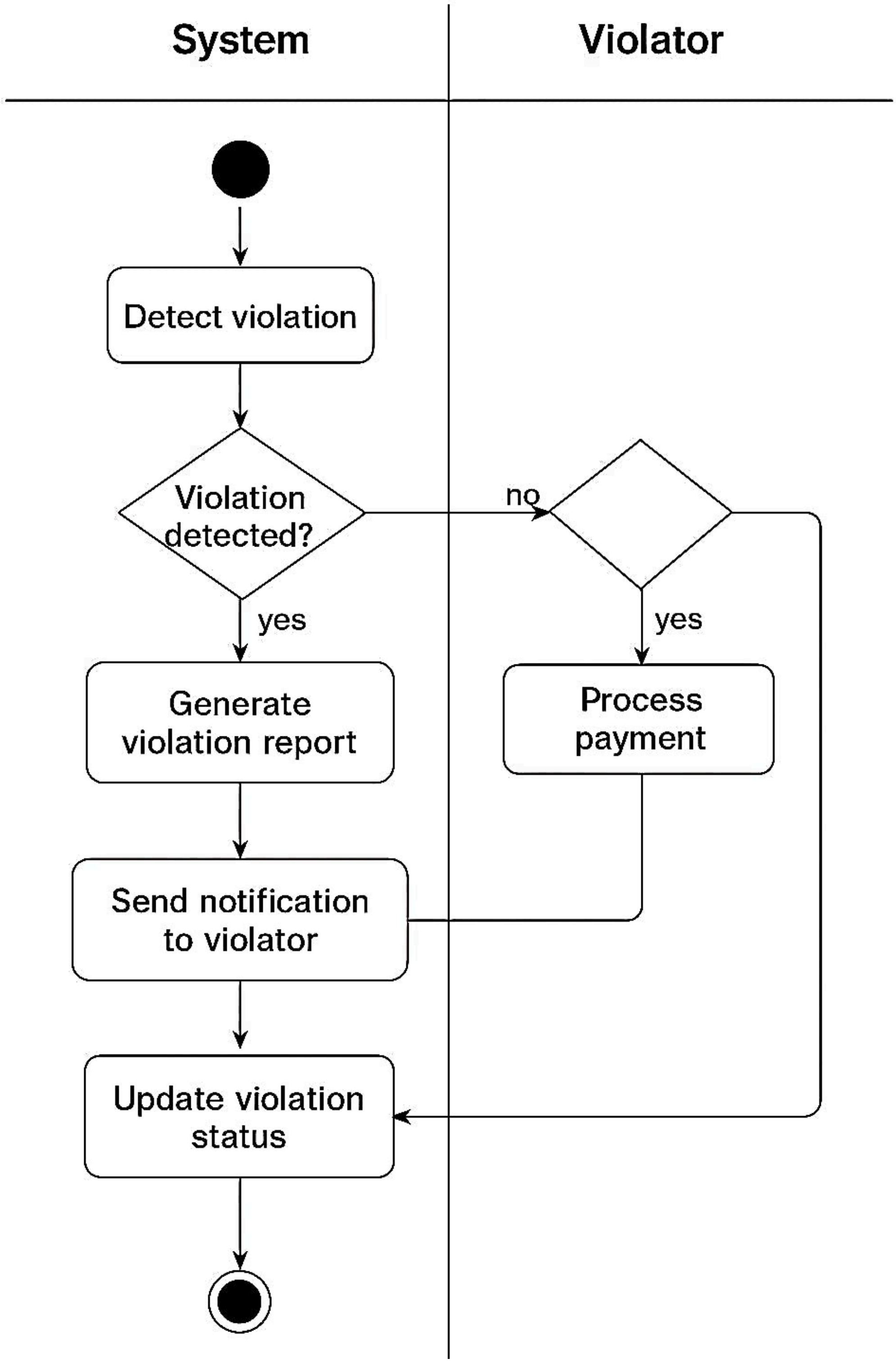
4.4 Deployment Diagram



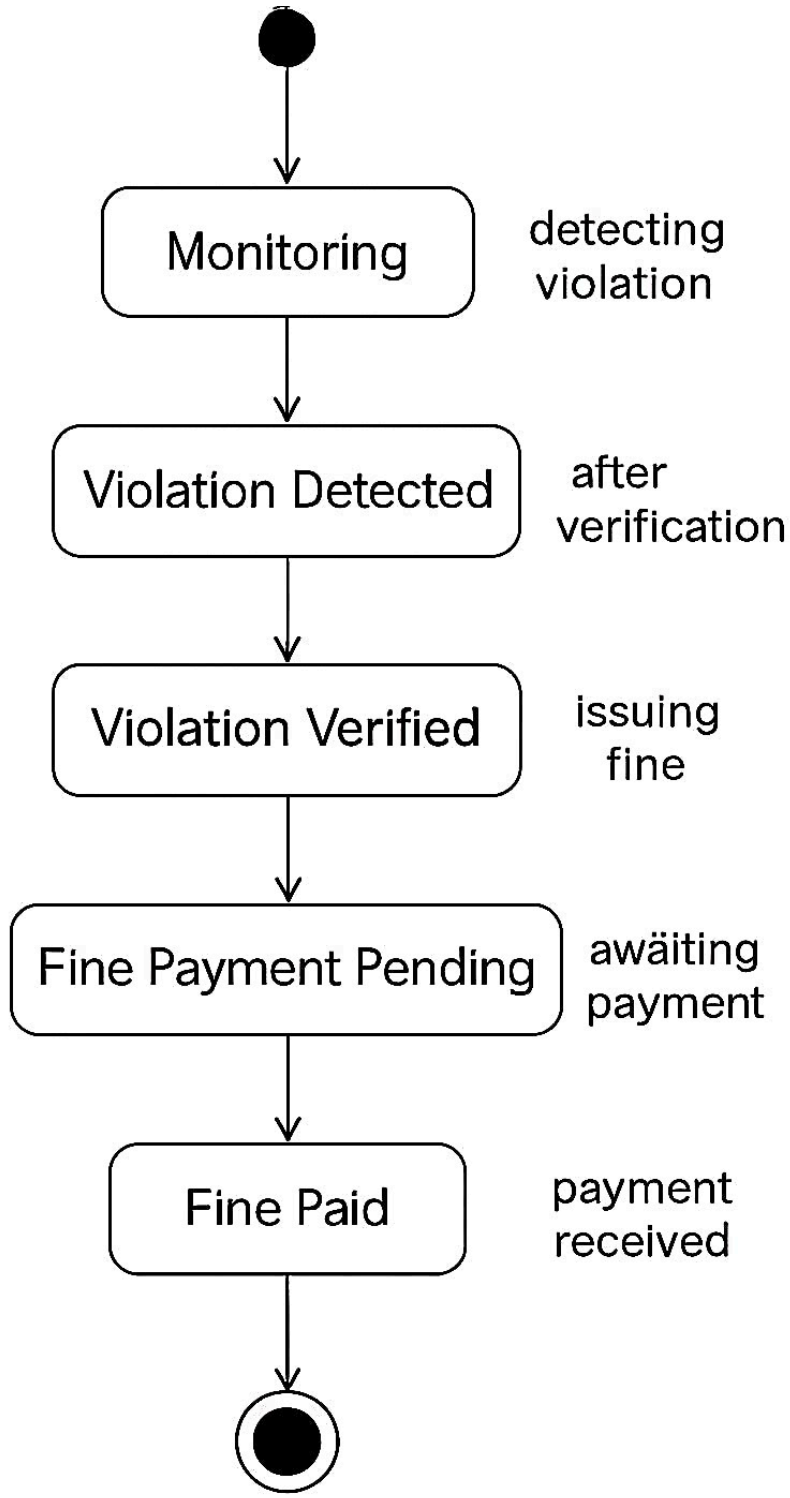
4.5 Use Case Diagram



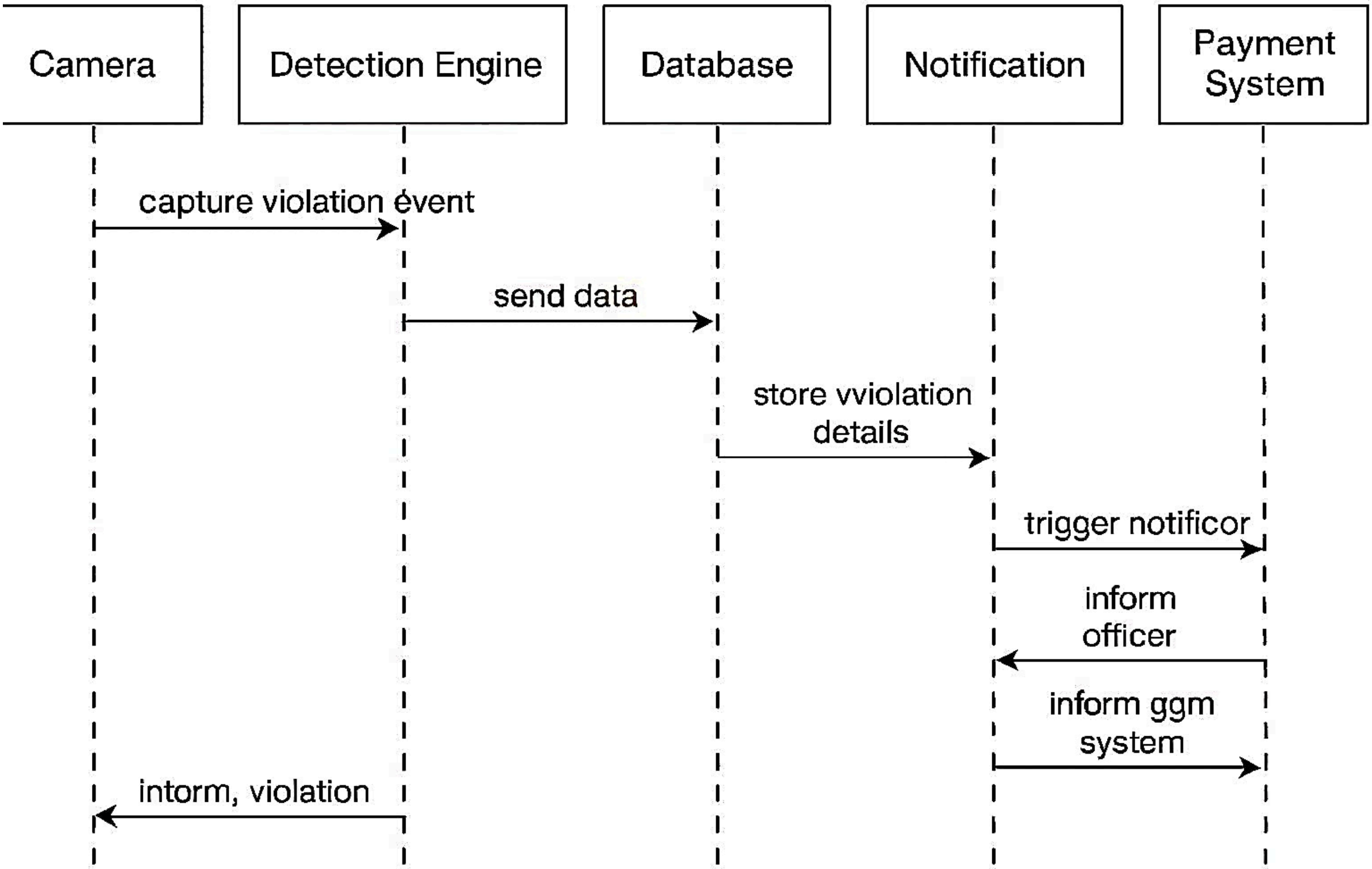
4.6 Activity Diagram



4.7 State Chart Diagram



4.8 Sequence Diagram



4.9 Collaborative Diagram